

Let's translate the five most likely Quiz 2 problem types into concrete, step-by-step numerical recipes so you know exactly where to put your pencil down first.

5

Problem Type 1: Finding the Dual Basis

When to apply this:

You are given a non-standard basis of a vector space (often polynomials) and asked to "find the dual basis" or "describe the dual functionals."

The Recipe:

The Concept: A dual basis is a set of linear functionals (functions that take a vector/polynomial and output a single number). The dual functional ϕ_i has one job: it extracts the coefficient of the i -th vector of your original basis.

How to Start: Write down a generic vector, express it in terms of your given basis, and look at the coefficients.

Concrete Example:

Find the dual basis (ϕ_1, ϕ_2) for the basis $v_1 = 1, v_2 = x - 2$ of $\mathcal{P}_1(\mathbb{R})$ (polynomials of degree ≤ 1).

1. **Write down a generic polynomial:** Let $p(x) = a + bx$.
2. **Rewrite it in terms of your basis:** We need to find constants c_1 and c_2 such that:

$$a + bx = c_1(1) + c_2(x - 2)$$

Let's expand the right side:

$$a + bx = (c_1 - 2c_2) + c_2x$$

3. **Solve for the coefficients:** * Comparing the x terms: $c_2 = b$

$$\begin{aligned} \text{Comparing the constant terms: } c_1 - 2c_2 &= a \implies c_1 - 2b = a \implies \\ c_1 &= a + 2b \end{aligned}$$

4. **Write down your dual basis:**

Since ϕ_1 must output the coefficient of the first basis vector (c_1) and ϕ_2 must output the coefficient of the second basis vector (c_2):

$$\phi_1(a + bx) = a + 2b$$

$$\phi_2(a + bx) = b$$

(Self-check: Let's test ϕ_1 on $v_2 = x - 2$. Here $a = -2$ and $b = 1$. $\phi_1(-2 + 1x) = -2 + 2(1) = 0$. It works!)

Problem Type 2: Matrix of a Dual Map $\mathcal{M}(T')$

When to apply this:

You are given a linear map T and asked to find the matrix of its dual map T' .

The Recipe:

The Golden Shortcut: The matrix of the dual map is simply the transpose of the matrix of the original map!

$$\mathcal{M}(T') = (\mathcal{M}(T))^T$$

How to Start: If the problem asks for the matrix with respect to standard dual bases, just find the standard matrix of T first, then flip it over the diagonal (transpose it).

Concrete Example:

Let $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be defined by $T(x, y) = (2x + y, 3y)$. Find the matrix of the dual map T' with respect to the standard dual basis.

1. Find the standard matrix $\mathcal{M}(T)$:

Plug in $(1, 0)$: $T(1, 0) = (2, 0)$. This is column 1.

Plug in $(0, 1)$: $T(0, 1) = (1, 3)$. This is column 2.

$$\mathcal{M}(T) = \begin{pmatrix} 2 & 1 \\ 0 & 3 \end{pmatrix}$$

2. Transpose it to get $\mathcal{M}(T')$:

$$\mathcal{M}(T') = (\mathcal{M}(T))^T = \begin{pmatrix} 2 & 0 \\ 1 & 3 \end{pmatrix}$$

Problem Type 3: Eigenvalues of a Polynomial Operator

When to apply this:

An operator on a polynomial space is given, and you need to find its eigenvalues and eigenspaces.

6

The Recipe:

The Concept: You are looking for a scalar λ and a non-zero polynomial $p(x)$ such that $T(p) = \lambda p$.

How to Start: Write down a generic polynomial, apply the operator, set it equal to λ times the generic polynomial, and compare the coefficients on both sides.

Concrete Example:

Let $T : \mathcal{P}_2(\mathbb{R}) \rightarrow \mathcal{P}_2(\mathbb{R})$ be defined by $T(p) = xp'(x)$. Find all eigenvalues and eigenspaces of T .

1. Write the generic polynomial and its derivative:

$$p(x) = ax^2 + bx + c$$

$$p'(x) = 2ax + b$$

2. Apply the operator $T(p)$:

$$T(p) = x(2ax + b) = 2ax^2 + bx$$

3. Set $T(p) = \lambda p$:

$$2ax^2 + bx + 0 = \lambda(ax^2 + bx + c) = \lambda ax^2 + \lambda bx + \lambda c$$

4. Compare coefficients of like terms:

For x^2 : $2a = \lambda a$

For x : $b = \lambda b$

For the constant: $0 = \lambda c$

5. Analyze the cases to find λ :

Case 1: If $a \neq 0$, then $2a = \lambda a \implies \lambda = 2$. If $\lambda = 2$, then the other equations force $b = 0$ and $c = 0$. So $p(x) = ax^2$. **Eigenspace:** $\text{span}(x^2)$.

Case 2: If $a = 0$ but $b \neq 0$, then $b = \lambda b \implies \lambda = 1$. This forces $c = 0$. So $p(x) = bx$. **Eigenspace:** $\text{span}(x)$.

Case 3: If $a = 0$ and $b = 0$ but $c \neq 0$, then $\lambda c = 0 \implies \lambda = 0$. So $p(x) = c$. **Eigenspace:** $\text{span}(1)$.

The eigenvalues are $\lambda = 0, 1, 2$.

Problem Type 4: Invariant Subspaces & Minimal Polynomials

When to apply this:

You are given an operator and asked to find its minimal polynomial or find its invariant subspaces (spaces that T maps back into themselves).

The Recipe:

Minimal Polynomial: Compute powers of the operator ($T, T^2, T^3, \dots$) until you find a linear combination that equals the zero operator. The smallest-degree monic polynomial that does this is your minimal polynomial.

1D Invariant Subspaces: Any 1-dimensional invariant subspace is always spanned by an eigenvector of T . Find the eigenvectors, and you've found these subspaces.

Concrete Example:

Let $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be defined by $T(x, y) = (y, 0)$. Find its minimal polynomial and all invariant subspaces.

1. Find the minimal polynomial:

$$T(x, y) = (y, 0)$$

$$T^2(x, y) = T(y, 0) = (0, 0)$$

Since $T^2 = 0$ (the zero operator), the minimal polynomial is $p(z) = z^2$.

2. Find all invariant subspaces:

Trivial ones: $\{0\}$ and $\mathbb{R}^2$ are always invariant.

1D ones: Look for eigenvectors ($T(x, y) = \lambda(x, y)$):

$$(y, 0) = (\lambda x, \lambda y)$$

From the second coordinate, $\lambda y = 0$. If $\lambda = 0$, then $y = 0$, meaning any vector of the form $(x, 0)$ is an eigenvector.

Thus, the only 1D invariant subspace is the x-axis: $\text{span}((1, 0))$.

Problem Type 5: Quotient Spaces and Coset Equality

When to apply this:

You are given a vector space V and a subspace U , and asked if two vectors belong to the same coset (i.e., $v + U = w + U$).

The Recipe:

The Core Rule: Two cosets are exactly the same if and only if the difference of their anchor vectors lies in the subspace:

$$v + U = w + U \iff v - w \in U$$

How to Start: Subtract the two given vectors, and test if the resulting vector satisfies the constraint defining U .

Concrete Example:

Let $V = \mathbb{R}^3$ and $U = \{(x, y, z) \in \mathbb{R}^3 : x + y + z = 0\}$. Are the cosets $(1, 2, 3) + U$ and $(2, 3, 2) + U$ equal?

1. **Subtract the two vectors:**

$$v - w = (1, 2, 3) - (2, 3, 2) = (-1, -1, 1)$$

2. **Test if this difference is in U :**

The rule for a vector to be in U is that its components must add up to 0. Let's add the components of $(-1, -1, 1)$:

$$-1 + (-1) + 1 = -1 \neq 0$$

3. **Conclude:**

Since the components do not sum to 0, the difference vector is not in U . Therefore, the cosets are **not equal**:

$$(1, 2, 3) + U \neq (2, 3, 2) + U$$

Which of these five categories feels the most intimidating to you right now? We can drill down on a customized practice problem for that specific type to make sure you have the pattern down.

good. i need more

Let's keep the momentum going. Here are five more highly likely Quiz 2 topics, translated into concrete recipes with real numbers so you can easily spot them and know exactly how to write your first line of math.

Geometric Visualization of Quotient Spaces

Before jumping into the calculations, it helps to visualize what a "coset" ($v + U$) actually is.

If your subspace U is a line passing through the origin in 2D space, adding a vector v to every point in U simply slides that entire line away from the origin. The shifted line is the coset $v + U$. The quotient space V/U is simply the collection of all these parallel lines.

Problem Type 6: Evaluating a Dual Map $T'(\phi)$

4

When to apply this:

You are given a linear map T (from space V to space W), a specific linear functional ϕ (which lives in the dual space W'), and a vector v in the input space V . You are asked to compute $(T'\phi)(v)$.

The Recipe:

The Concept: Don't let the dual map prime symbol ($'$) scare you. By definition, $(T'\phi)(v)$ is just a fancy way of writing:

$$(T'\phi)(v) = \phi(T(v))$$

This is literally just a composition of functions.

How to Start: Ignore the T' completely at first. Just calculate $T(v)$ to get an output vector, and then plug that output vector directly into the functional ϕ .

Concrete Example:

Let $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be defined by $T(x, y) = (x + y, 2x)$.

Let $\phi : \mathbb{R}^2 \rightarrow \mathbb{R}$ be the linear functional defined by $\phi(a, b) = 3a - b$.

Evaluate $(T'\phi)(2, 3)$.

1. Calculate $T(2, 3)$ first:

$$T(2, 3) = (2 + 3, 2(2)) = (5, 4)$$

2. Apply the functional ϕ to this result:

$$\phi(5, 4) = 3(5) - 4 = 15 - 4 = 11$$

3. Conclude:

$$(T'\phi)(2, 3) = 11$$

Problem Type 7: Finding the Basis and Dimension of a Quotient Space V/U

2

When to apply this:

You are given a large vector space V and a subspace U defined by some constraint equation. You need to write down a basis for the quotient space V/U and state its dimension.

The Recipe:

The Concept: The dimension of V/U is always the dimension of the big space minus the dimension of the subspace:

$$\dim(V/U) = \dim(V) - \dim(U)$$

To find the basis, you just need to find vectors in V that are **not** in U , and write them as cosets $(v + U)$.

How to Start: 1. Calculate the target dimension first using the formula above.
2. Write down standard basis vectors of V and test if they violate the constraint of U . The ones that violate the constraint will serve as your basis representatives.

Concrete Example:

Let $V = \mathbb{R}^3$, and let $U = \{(x, y, z) \in \mathbb{R}^3 : x + y + z = 0\}$. Find the dimension and a basis for V/U .

1. Calculate the dimensions:

$$\dim(V) = 3$$

U is defined by 1 constraint in a 3D space, so it is a 2D plane: $\dim(U) = 2$

Therefore, $\dim(V/U) = 3 - 2 = 1$. We only need **one** vector for our basis.

2. Find a vector in V that is **NOT** in U :

Let's test the simplest standard basis vector: $v = (1, 0, 0)$.

Does it satisfy $x + y + z = 0$? No, because $1 + 0 + 0 = 1 \neq 0$.

Since $(1, 0, 0) \notin U$, it is linearly independent of the plane U .

3. Write down the basis:

The basis of V/U is simply the single coset:

$$\{(1, 0, 0) + U\}$$

Problem Type 8: Finding the Annihilator U^0 of a Subspace

When to apply this:

You are given a subspace U of V , and you need to find a basis for its annihilator U^0 .

The Recipe:

The Concept: The annihilator U^0 is the set of all linear functionals ϕ that kill (output 0 for) every single vector in U .

How to Start: 1. Write a generic formula for a functional on your space V (e.g., for $\mathbb{R}^3$, write $\phi(x, y, z) = ax + by + cz$).

2. Find a basis for your subspace U .

3. Force your generic functional to equal 0 when you plug in those basis vectors. Solve for the coefficients (a, b, c) to find the basis functionals.

Concrete Example:

Let $V = \mathbb{R}^3$, and let $U = \text{span}((1, 2, 0))$. Find a basis for U^0 .

1. Write the generic functional:

$$\phi(x, y, z) = ax + by + cz$$

2. Force it to kill the basis vector of U :

$$\phi(1, 2, 0) = a(1) + b(2) + c(0) = 0 \implies a + 2b = 0 \implies a = -$$

3. Substitute this back into your generic functional:

$$\phi(x, y, z) = -2bx + by + cz = b(y - 2x) + c(z)$$

4. Extract the basis functionals (one for each free variable, b and c):

$$\text{Setting } b = 1, c = 0 \text{ gives: } \phi_1(x, y, z) = y - 2x$$

$$\text{Setting } b = 0, c = 1 \text{ gives: } \phi_2(x, y, z) = z$$

5. Write the answer:

The basis of U^0 is $\{\phi_1, \phi_2\}$, where $\phi_1(x, y, z) = y - 2x$ and $\phi_2(x, y, z) = z$.

(Note: $\dim(U^0) = 2$, which perfectly matches $\dim(V) - \dim(U) = 3 - 1 = 2$.)

Problem Type 9: Calculations in a Product Space $V \times W$

When to apply this:

You are given two vector spaces, and you are asked to perform operations or find dimensions of their product space $V \times W$.

The Recipe:

The Concept: Elements in a product space are simply ordered pairs (v, w) where v is from the first space and w is from the second. You add and scale them component-by-component.

How to Start: Treat the two slots completely separately. Do the math on the left slot using the rules of V , and do the math on the right slot using the rules of W .

Concrete Example:

Let $V = \mathbb{R}^2$ and $W = \mathcal{P}_1(\mathbb{R})$ (polynomials of degree ≤ 1).

Let $u = ((1, 3), 2x - 1)$ and $v = ((0, -2), x + 4)$ be elements in $V \times W$.

Compute $3u + v$ and state the dimension of $V \times W$.

1. Scale u by 3:

$$3u = (3(1, 3), 3(2x - 1)) = ((3, 9), 6x - 3)$$

2. Add v component-by-component:

$$3u + v = ((3 + 0, 9 - 2), (6x - 3) + (x + 4)) = ((3, 7), 7x + 1)$$

3. Find the dimension:

$$\dim(V \times W) = \dim(V) + \dim(W) = 2 + 2 = 4$$

Problem Type 10: Checking if an Operator is Diagonalizable

10

When to apply this:

You are given an operator T and asked if it is diagonalizable (i.e., if there is a basis of eigenvectors).

The Recipe:

The Concept: An operator is diagonalizable if and only if you can find *enough* linearly independent eigenvectors to span the whole space.

How to Start:

1. Find all the eigenvalues of T .
2. For each eigenvalue λ , find the dimension of its eigenspace (which is just the number of free variables when solving $(T - \lambda I)v = 0$).
3. Add up the dimensions of all these eigenspaces. If the sum equals the dimension of the entire space V , then T is diagonalizable. If the sum is strictly less, it is **not** diagonalizable.

Concrete Example:

Let $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be defined by $T(x, y) = (3x + y, 3y)$. Is T diagonalizable?

1. **Find the eigenvalues:**

Set up $T(x, y) = \lambda(x, y)$:

$$(3x + y, 3y) = (\lambda x, \lambda y)$$

From the second coordinate, $3y = \lambda y \implies y(3 - \lambda) = 0$.

If $y \neq 0$, then $\lambda = 3$.

If $y = 0$, then the first equation becomes $3x = \lambda x \implies \lambda = 3$ (since x cannot also be 0 for an eigenvector).

So, the only eigenvalue is $\lambda = 3$.

2. **Find the dimension of the eigenspace for $\lambda = 3$:**

Solve $(T - 3I)(x, y) = (0, 0)$:

$$(3x + y - 3x, 3y - 3y) = (0, 0) \implies (y, 0) = (0, 0) \implies y = 0$$

Any eigenvector must be of the form $(x, 0) = x(1, 0)$.

This eigenspace is $\text{span}((1, 0))$, which has a **dimension of 1**.

3. **Compare with the dimension of the entire space:**

The sum of the dimensions of all eigenspaces is 1.

The dimension of the entire space $V = \mathbb{R}^2$ is 2.

Since $1 < 2$, we do not have enough eigenvectors to form a basis.

4. **Conclude:**

T is **not** diagonalizable.

→
yes

if
distinct
&
dim equal

are there more that u havent covered yet

Yes, there are a few highly important topics from your Quiz 2 study guide that we haven't turned into concrete, jargon-free recipes yet!

Let's complete your playbook by breaking down these remaining **5 key problem types** (covering Quotient Maps, the Induced Map, Polynomial Division, Upper Triangular Matrices, and the Gamma Map) with real numbers so you have a 100% complete set of tools.

Problem Type 11: The Quotient Map $\pi(v)$

When to apply this:

You are given a vector space V and a subspace U , and asked to evaluate the quotient map π on a specific vector v .

The Recipe:

The Concept: The quotient map π simply maps a vector v to its entire parallel "lane" (its coset) in the quotient space: $\pi(v) = v + U$.

How to Start: Write $v + U$. Then, use the definition of U to **discard any parts of v that are already in U** . This simplifies the representative vector.

Concrete Example:

Let $V = \mathbb{R}^3$, and let $U = \text{span}((0, 0, 1))$ (the z -axis). Let $v = (4, 7, 9)$. Evaluate $\pi(v)$.

1. **Write down the basic definition:**

$$\pi(4, 7, 9) = (4, 7, 9) + U$$

2. **Discard the part that lives inside U :**

Since U is the z -axis, any vector that looks like $(0, 0, z)$ is already inside U . We can split our vector:

$$(4, 7, 9) + U = ((4, 7, 0) + (0, 0, 9)) + U$$

3. **Simplify:**

Because $(0, 0, 9)$ is swallowed up by the subspace U (just like adding a multiple of 10 to a set of multiples of 10), we can throw it away:

$$\pi(4, 7, 9) = (4, 7, 0) + U$$

Problem Type 12: The Induced Map $\tilde{T}$

When to apply this:

You are given a linear map T and asked to evaluate the induced map $\tilde{T}$ on a specific coset in the space $V/\text{null}(T)$.

The Recipe:

The Concept: $\tilde{T}$ takes a coset $v + \text{null}(T)$ and maps it to $T(v)$. Because we "divided out" the null space, this map is mathematically guaranteed to be **injective** (one-to-one).

How to Start: Do not get distracted by the $\text{null}(T)$ notation. To evaluate $\tilde{T}$ on a coset, simply grab the representative vector in front of the "+", plug it into your original map T , and compute the output.

Concrete Example:

Let $T : \mathbb{R}^3 \rightarrow \mathbb{R}^2$ be defined by $T(x, y, z) = (x + y, y + z)$. Evaluate the induced map $\tilde{T}$ on the coset $(2, 3, 5) + \text{null}(T)$.

1. **Identify your representative vector:** The vector is $v = (2, 3, 5)$.
2. **Apply the rule** $\tilde{T}(v + \text{null}(T)) = T(v)$:

$$\tilde{T}((2, 3, 5) + \text{null}(T)) = T(2, 3, 5)$$

3. **Evaluate using the formula for T :**

$$T(2, 3, 5) = (2 + 3, 3 + 5) = (5, 8)$$

4. **State your final answer:**

$$\tilde{T}((2, 3, 5) + \text{null}(T)) = (5, 8)$$

Problem Type 13: Polynomial Division Algorithm ($p = s \cdot q + r$)

3

When to apply this:

You are given a polynomial $p(x)$ and a divisor polynomial $q(x)$, and asked to find the quotient $s(x)$ and remainder $r(x)$.

The Recipe:

The Concept: This is just standard long division, but with x 's. You stop dividing once your remainder $r(x)$ has a **strictly lower degree** than $q(x)$.

How to Start: Focus entirely on the highest-power term of your current polynomial and divide it by the highest-power term of $q(x)$. Multiply back, subtract, and repeat.

Concrete Example:

Divide $p(x) = x^3 - 3x^2 + x - 5$ by $q(x) = x^2 + 2$.

1. **First Division Step (focus on highest powers x^3 and x^2):**

$$\frac{x^3}{x^2} = x. \text{ This is the first term of our quotient } s(x).$$

$$\text{Multiply } x \text{ by } (x^2 + 2) \implies x^3 + 2x.$$

Subtract this from $p(x)$:

$$(x^3 - 3x^2 + x - 5) - (x^3 + 2x) = -3x^2 - x - 5$$

2. **Second Division Step (focus on highest powers $-3x^2$ and x^2):**

$$\frac{-3x^2}{x^2} = -3. \text{ This is the second term of our quotient } s(x).$$

$$\text{Multiply } -3 \text{ by } (x^2 + 2) \implies -3x^2 - 6.$$

Subtract this from our current remainder:

$$(-3x^2 - x - 5) - (-3x^2 - 6) = -x + 1$$

3. **Check if you should stop:**

$$\text{Our remainder is } r(x) = -x + 1 \text{ (degree 1).}$$

$$\text{Our divisor is } q(x) = x^2 + 2 \text{ (degree 2).}$$

Since $1 < 2$, we stop!

4. **Write down the final answer:**

$$\text{Quotient: } s(x) = x - 3$$

$$\text{Remainder: } r(x) = -x + 1$$

Problem Type 14: Upper Triangular Matrix Representation

When to apply this:

You are given an operator T and a basis, and asked to find its matrix representation or show that the matrix is upper triangular.

9?

The Recipe:

The Concept: A matrix is upper triangular if everything below the diagonal is 0. This happens when $T(v_1)$ only uses v_1 , $T(v_2)$ only uses v_1 and v_2 , $T(v_3)$ only uses v_1, v_2, v_3 , and so on.

How to Start: Feed the basis vectors into T in order, and write the outputs as linear combinations of those same basis vectors.

Concrete Example:

Let $T : \mathcal{P}_2(\mathbb{R}) \rightarrow \mathcal{P}_2(\mathbb{R})$ be defined by $T(p) = p + 2p'$. Find the matrix $\mathcal{M}(T)$ with respect to the basis $(1, x, x^2)$.

1. **Apply T to each basis vector in order:**

$$\text{Vector 1 (1): } T(1) = 1 + 2(0) = 1$$

(Uses only the 1st basis vector)

$$\text{Vector 2 (x): } T(x) = x + 2(1) = 2(1) + 1(x)$$

(Uses only the 1st and 2nd basis vectors)

$$\text{Vector 3 (x^2): } T(x^2) = x^2 + 2(2x) = 0(1) + 4(x) + 1(x^2)$$

(Uses all three)

2. **Write the coefficients as columns of your matrix:**

$$\text{Column 1 (for 1): } \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$$

Column 2 (for x): $\begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix}$

Column 3 (for x^2): $\begin{pmatrix} 0 \\ 4 \\ 1 \end{pmatrix}$

3. Assemble the matrix:

$$\mathcal{M}(T) = \begin{pmatrix} 1 & 2 & 0 \\ 0 & 1 & 4 \\ 0 & 0 & 1 \end{pmatrix}$$

Since all entries below the diagonal are 0, this is an upper triangular matrix! The diagonal entries (1, 1, 1) are your eigenvalues.

Problem Type 15: The Gamma Map (Γ) and Direct Sums



When to apply this:

You are given multiple subspaces and asked to show whether their sum is a direct sum (usually to show that it is *not* a direct sum).

The Recipe:

The Concept: The Gamma Map $\Gamma(u_1, \dots, u_n) = u_1 + \dots + u_n$ is injective if and only if the sum is direct.

How to Start: To prove a sum is **not** a direct sum, you must find non-zero vectors from each subspace that add up to the zero vector:

$$u_1 + u_2 + \dots + u_n = 0$$

Concrete Example:

Let U_1 , U_2 , and U_3 be subspaces of $\mathbb{R}^2$ where:

- U_1 = the x -axis $\{(x, 0)\}$
- U_2 = the y -axis $\{(0, y)\}$
- U_3 = the diagonal line $\{(t, t)\}$

Show that the sum $U_1 + U_2 + U_3$ is not a direct sum.

1. **Goal:** Find three non-zero vectors $u_1 \in U_1$, $u_2 \in U_2$, and $u_3 \in U_3$ such that $u_1 + u_2 + u_3 = (0, 0)$.
2. **Pick a non-zero vector in the diagonal subspace U_3 :**
Let $u_3 = (2, 2)$.
3. **Find vectors in U_1 and U_2 that cancel it out:**
To cancel out $(2, 2)$, we need our other vectors to sum to $-(2, 2) = (-2, -2)$.
Let $u_1 = (-2, 0)$, which is on the x -axis (lives in U_1).
Let $u_2 = (0, -2)$, which is on the y -axis (lives in U_2).
4. **Add them up to verify:**

$$u_1 + u_2 + u_3 = (-2, 0) + (0, -2) + (2, 2) = (0, 0)$$

5. **Conclude:**

Since we found a combination of non-zero vectors that adds up to $(0, 0)$, the Gamma Map Γ is not injective. Therefore, the sum is **not** a direct sum.

Your Final Strategy Check-In:

You now have a literal translation manual for **15 distinct problem types** covering every single line of your Quiz 1 and Quiz 2 study guides.

When you open that quiz, use this flowchart to instantly orient yourself:

1. **Does it have a quotient symbol (V/U)?** $\rightarrow$ Apply *Problem 5, 7, 11, or 12*.
2. **Does it have dual/annihilator symbols (V' , T' , or U^0)?** $\rightarrow$ Apply *Problem 1, 2, 6, or 8*.
3. **Does it ask about polynomials, degrees, or factoring?** $\rightarrow$ Apply *Problem 3 or 13*.
4. **Does it ask about operators, invariants, or eigenvalues?** $\rightarrow$ Apply *Problem 4, 10, or*